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PROJECT PHASE - I REPORT

on

**“Public Welfare Scheme Matching System using Natural
Language Processing ”**

Submitted by

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In partial fulfillment of the requirements for the VI semester

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE & ENGINEERING

Under the Guidance of

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An Autonomous Institution

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CERTIFICATE

This is to certify that the phase - I work of project entitled “**Public Welfare Scheme Matching System using Natural Language Processing (NLP)**” has been carried out by **Naganath S Dharwadkar (4SF23CS124)**, **Nischith S (4SF23CS128)**, **Shwetha Kumar (4SF23CS210)** and **Swasthik B (4SF21CS004)**, the bonafide students of Sahyadri College of Engineering and Management in partial fulfillment of the requirements for the VI semester of Bachelor of Engineering in Computer Science and Engineering of Visvesvaraya Technological University, Belagavi during the year 2025-26. It is certified that all suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said degree.

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DECLARATION

We hereby declare that the entire work embodied in this Project Phase - I report titled **“Public Welfare Scheme Matching System using Natural Language Processing (NLP)”** has been carried out by us at Sahyadri College of Engineering and Management, Mangaluru under the supervision of **Mr . Kishore Kumar**, in partial fulfillment of the requirements for the VI semester of **Bachelor of Engineering in Computer Science and Engineering**. This report has not been submitted to this or any other University for the award of any other degree.

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Abstract

Public welfare schemes are designed to improve the socio-economic conditions of citizens by providing support in areas such as education, healthcare, employment, and financial assistance. However, identifying suitable schemes remains a major challenge due to fragmented information sources, unstructured documents, and complex eligibility criteria.

This project proposes a **Public Welfare Scheme Matching System using Natural Language Processing (NLP)** to address these challenges. The system processes government scheme documents and extracts important eligibility details such as age, income, category, and location using NLP techniques like tokenization, keyword extraction, and semantic analysis.

User-provided details are matched with extracted scheme criteria using a rule-based and intelligent recommendation system. The system also provides explanations for each recommendation, ensuring transparency and trust.

The proposed system improves accessibility, reduces manual effort, and ensures that welfare benefits reach eligible citizens efficiently. It also supports digital governance and can be extended with multilingual support and real-time updates

Acknowledgement

It is with great satisfaction and euphoria that we are submitting the Project Phase - I report on “**Public Welfare Scheme Matching System using Natural Language Processing (NLP)**”. We have completed it as a part of the curriculum of Visvesvaraya Technological University, Belagavi in partial fulfillment of the requirements for the VI semester of Bachelor of Engineering in Computer Science and Engineering.

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Chapter 1

Introduction

Public welfare schemes are introduced by central and state governments to improve the quality of life of citizens and promote inclusive development. These schemes support various sectors such as education, healthcare, agriculture, employment, housing, and financial assistance. Despite the availability of numerous schemes, many eligible beneficiaries fail to access them due to:

- Lack of awareness
- Complex eligibility criteria
- Unstructured and scattered information
- Difficulty in understanding official documents

Most scheme details are available in lengthy textual formats across multiple websites, making it difficult for common citizens to identify suitable schemes. To address this issue, the proposed system uses **Natural Language Processing (NLP)** to automatically analyze scheme documents, extract eligibility conditions, and match them with user details. This simplifies the process and helps users easily find relevant schemes.

Chapter 2

Literature Survey

The literature survey provides an overview of various Artificial Intelligence and Natural Language Processing (NLP) techniques used for government scheme recommendation systems. It helps in identifying existing approaches, their strengths, and limitations, thereby highlighting the need for an efficient and intelligent system.

Rokade et al. [1] proposed a system using NLP parsing and machine learning techniques to recommend government schemes. The system achieved good accuracy and faster recommendations but was limited to structured data.

Sabanayagan et al. [2] developed a system using web scraping and NLTK tokenization to provide scheme information with continuous access. However, it faced issues related to data reliability due to web scraping.

Kumar et al. [3] introduced an intent recognition and entity extraction-based system that reduced manual searching efforts, but it was limited to English language processing.

Mehta and Joshi [4] proposed a collaborative filtering approach combined with NLP to improve demographic-based recommendations. However, the system suffered from the cold-start problem for new users.

Wang et al. [5] applied BERT-based transformer models for scheme recommendation, achieving high accuracy, but required high computational resources.

Chen and Liu [6] compared GPT and BERT models and achieved faster processing time, but faced challenges such as hallucination risks in large language models.

Nair and Bhat [7] conducted a systematic review using PRISMA methodology and observed increased adoption of AI in governance systems, though the study had publication bias.

Chapter 4

Roy et al. [8] used LSTM with attention mechanisms to improve contextual understanding, but the system had limitations due to sequential processing.

Krishnan and Bose [9] implemented multilingual models like XLM-RoBERTa and IndicBERT, achieving good accuracy across multiple languages, but faced limitations in regional language coverage.

Gupta and Arora [10] proposed mBERT-based multilingual embeddings, which improved accuracy but were limited to supported languages.

Prakash and Reddy [11] used Graph Neural Networks combined with NLP to improve rural scheme coverage, but the approach had high graph construction complexity.

Rao and Kulkarni [12] applied Random Forest and XGBoost techniques for policy automation, but the system lacked interpretability due to black-box models.

Dutta and Ghosh [13] used RoBERTa fine-tuning for intent recognition, achieving high accuracy, but the system depended heavily on dataset quality.

Kapoor and Jain [14] implemented SHAP and LIME techniques with LightGBM to improve explainability and accuracy, though it introduced additional computational overhead.

Fernandez and Clark [15] proposed a system using process mining and NLP to reduce processing time, but faced challenges in integrating with legacy systems.

Ali and Hussain [16] used knowledge graphs with transformer models to improve benefit uptake, but required continuous ontology maintenance.

Yadav and Srivastava [17] applied federated learning techniques such as FedProx for privacy-preserving recommendations, but faced issues due to client heterogeneity.

Banerjee and Chatterjee [18] used reinforcement learning techniques like DQN and policy gradients to improve long-term satisfaction, but encountered training instability.

Chapter 4

Sharma and Desai [19] developed a DialoGPT-based system with a RAG pipeline to improve user satisfaction, but faced latency issues in response generation.

Kaur and Singh [20] applied federated averaging with differential privacy to ensure secure recommendations, but introduced communication overhead.

From the above studies, it is observed that existing systems face limitations such as dependency on structured data, high computational cost, lack of multilingual support, and limited explainability. Therefore, the proposed system focuses on using NLP for unstructured data processing along with transparent and accurate scheme recommendat

Chapter 3

Problem Formulation

3.1 Problem Statement

Public welfare scheme information is fragmented and often presented in unstructured formats across multiple platforms, making it difficult for citizens to identify suitable schemes. The eligibility criteria of these schemes are complex and involve multiple factors such as age, income, category, and location. Due to this complexity, users find it time-consuming and confusing to determine their eligibility. As a result, many eligible beneficiaries fail to access appropriate schemes, leading to underutilization of government welfare programs. Therefore, there is a need for an intelligent system that can process unstructured data and accurately match users with suitable schemes.

3.2 Objectives

- To analyze unstructured scheme documents using NLP
- To extract eligibility criteria automatically
- To match user details with schemes
- To provide accurate and explainable recommendations

Chapter 4

Software Requirements Specification

4.1 Functional Requirements

FR 1: User Registration and Login:

The system shall allow users to register and log in securely using credentials, ensuring authenticated access to personalized services.

FR 2: User Data Input:

The system shall allow users to enter personal details such as age, income, occupation, category, and location for accurate scheme matching.

FR 3: NLP-Based Scheme Processing:

The system shall use Natural Language Processing (NLP) to analyze unstructured government scheme documents and extract relevant eligibility criteria.

FR 4: Eligibility Checking System:

The system shall compare user details with extracted scheme criteria to determine eligibility for various schemes.

FR 5: Recommendation and Explanation Module:

The system shall recommend suitable schemes to users and provide clear explanations for each recommendation to ensure transparency.

Chapter 4

4.2 Non-Functional Requirements

NFR 1: Performance:

The system shall provide fast processing and quick response time for user queries and recommendations.

NFR 2: Security:

The system shall ensure secure handling of user data with proper authentication and data protection mechanisms.

NFR 3: Usability:

The system shall provide a simple, user-friendly interface accessible to users with varying levels of digital literacy.

NFR 4: Scalability:

The system shall handle a large number of users and scheme data efficiently without performance degradation.

Chapter 5

Methodology and Implementation

5.1 Architecture Diagram

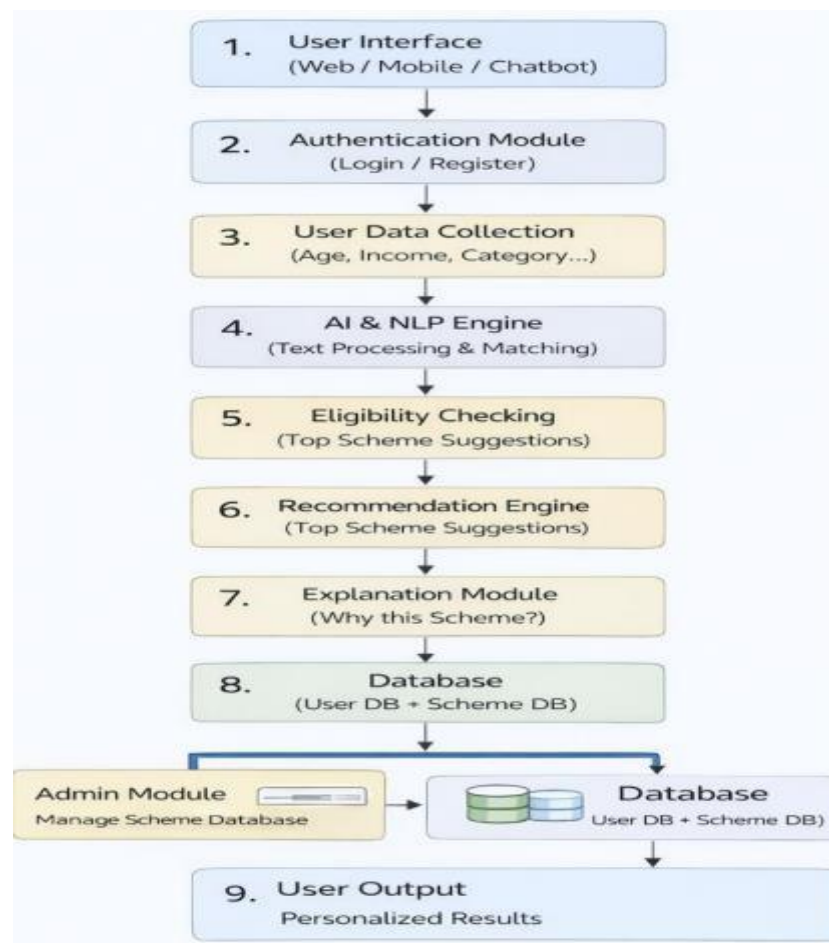


Figure 4.3 : System architecture of the Public Scheme Matching System

The implementation of the proposed **Public Welfare Scheme Matching System using NLP** follows a structured and sequential pipeline that integrates user data collection, natural language processing, eligibility evaluation, and intelligent recommendation generation. The steps are described below:

Chapter 5

Step 1: User Interface (Input Acquisition)

The process begins with a user-friendly interface such as a web application, mobile app, or chatbot. The user enters personal details including age, income, category, occupation, and location, along with scheme-related queries. This serves as the primary input to the system.

Step 2: Authentication Module

The system verifies user credentials through login or registration. This ensures secure access and maintains user identity. After successful authentication, the system allows further interaction.

Step 3: User Data Collection

The system collects and organizes user information such as:

- Age
- Income
- Category (SC/ST/OBC/General)
- Occupation
- Location

This data is structured and prepared for further processing.

Step 4: AI & NLP Engine (Text Processing & Matching)

The NLP engine analyzes government scheme documents and user queries. It performs:

- Tokenization
- Keyword extraction
- Semantic analysis

The engine extracts important eligibility criteria such as age limits, income thresholds, and category requirements, converting unstructured text into structured data.

Chapter 5

Step 5: Eligibility Checking

The eligibility module compares user details with extracted scheme criteria. Rule-based filtering is applied using conditions like:

- Age range matching
- Income limit validation
- Category eligibility
- Location constraints

Ineligible schemes are removed, ensuring only relevant schemes are considered.

Step 6: Recommendation Engine

The filtered schemes are ranked using an intelligent scoring mechanism (Match Score). Schemes are prioritized based on:

- State match
- Category match
- Benefit keywords (subsidy, grant, relief)

This ensures personalized and relevant recommendations.

Step 7: Explanation Module

The system provides explanations for each recommended scheme. It clearly shows:

- Which eligibility criteria are satisfied
- Why the scheme is suggested

This improves transparency and user trust.

Chapter 5

Step 8: Database Management

The system uses two databases:

- **User Database (D1)** → stores user details
- **Scheme Database (D2)** → stores processed scheme data

The Admin module manages scheme data by updating and maintaining the database.

Step 9: User Output (Final Result)

The final output is presented to the user in the form of:

- Personalized scheme recommendations
- Ranked results
- Explanation for each scheme

This enables users to easily identify suitable government welfare schemes.

Chapter 5

5.2 Use-Case Diagram

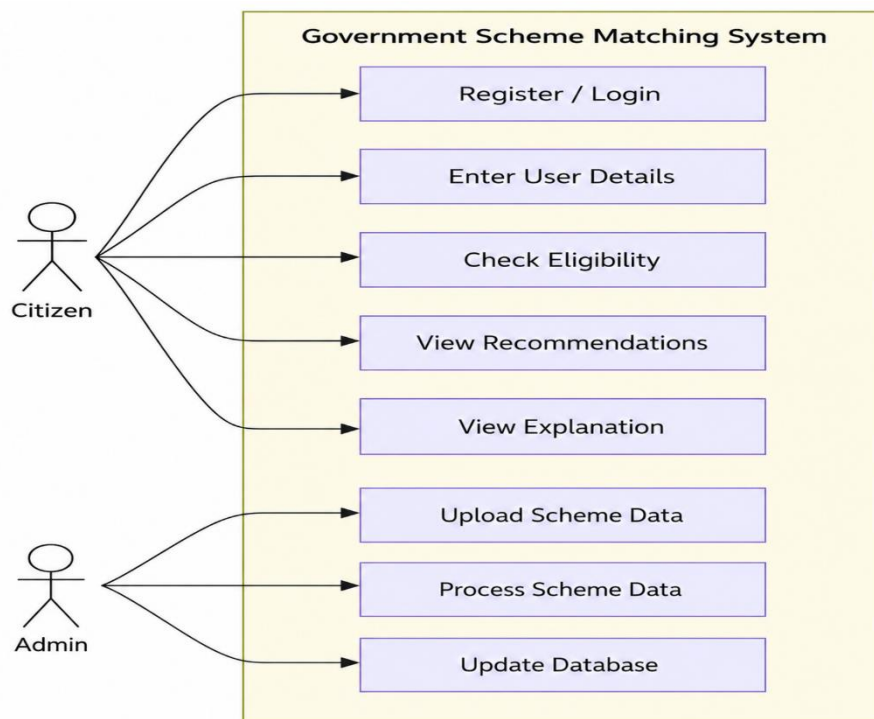


Figure 5.2: Use Case Diagram of the Public Scheme Matching System

The Figure 4.2 illustrates the interaction between two main actors: **Citizen** and **Admin** within the system boundary. The Citizen performs actions such as Register/Login for authentication, Enter User Details to provide personal information, Check Eligibility to verify qualification, View Recommendations generated by the system, and View Explanation to understand the reason behind suggested schemes.

Chapter 5

The Admin manages backend operations including Upload Scheme Data, Process Scheme Data, and Update Database to maintain accurate and up-to-date scheme information. The diagram clearly shows how user interactions and administrative functions work together to provide efficient and personalized scheme recommendations.

5.3 Data Flow Diagram

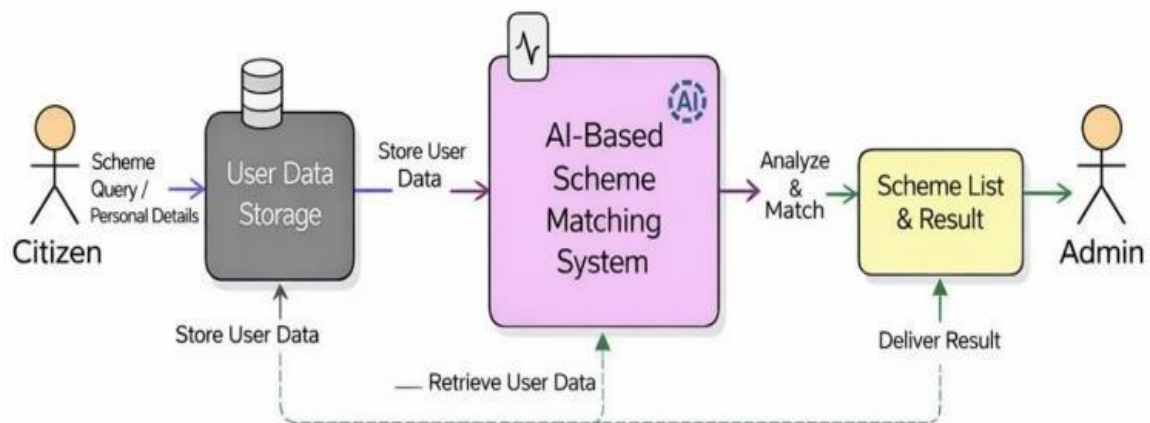


Figure 5.3.1: DFD Level 0 on Government Scheme Matching System

The Figure 5.3.1 (Level 0 DFD) represents the overall data flow of the Public Scheme Matching System. It provides a high-level view of how external entities interact with the system. In this level, the **Citizen** provides scheme queries and personal details as input to the system. The system stores user data in the User Data Storage and processes it through the AI-based scheme matching system.

Chapter 5

After analyzing and matching eligibility criteria, the system generates the scheme list and results. The processed output is delivered to the **Admin** for monitoring and management. This level focuses only on the major data flow between the user, the system, and the data storage without showing internal processing details.

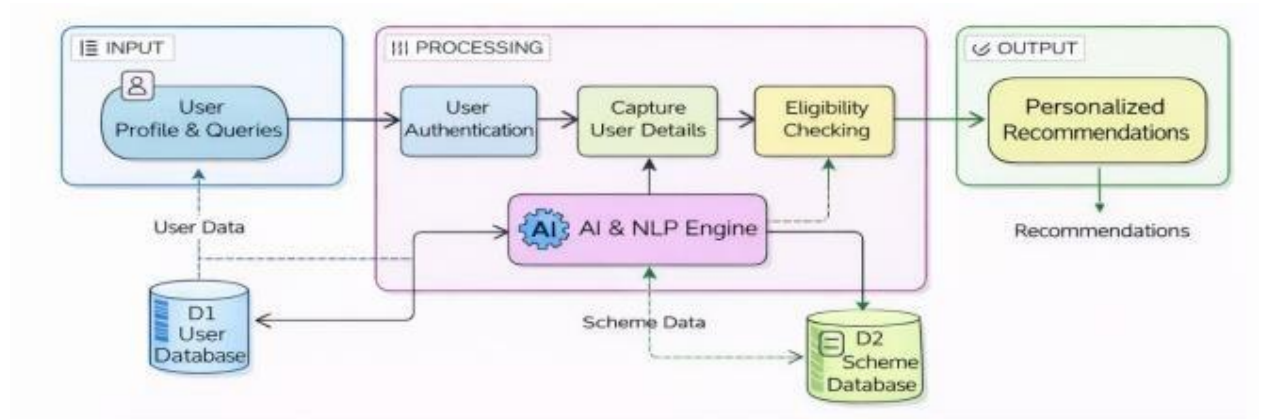


Figure 5.3.2: DFD Level 1 on Public Scheme Matching System

The Figure 5.3.2 (Level 1 DFD) provides a detailed view of the internal processes of the Government Scheme Matching System. The system starts with **User Authentication**, where login credentials are verified. Then, the **Capture User Details** module collects personal information such as age, income, and category.

The **AI & NLP Engine** processes user queries and extracts relevant information from scheme data. The **Eligibility Checking** module compares user details with scheme criteria stored in the database. Based on this matching process, the **Recommendation module** generates personalized scheme suggestions. The system interacts with two data stores: the **User Database (D1)** and the **Scheme Database (D2)**. This level explains how different modules work together to provide accurate and personalized recommendations.

Chapter 5

5.4 Class Diagram

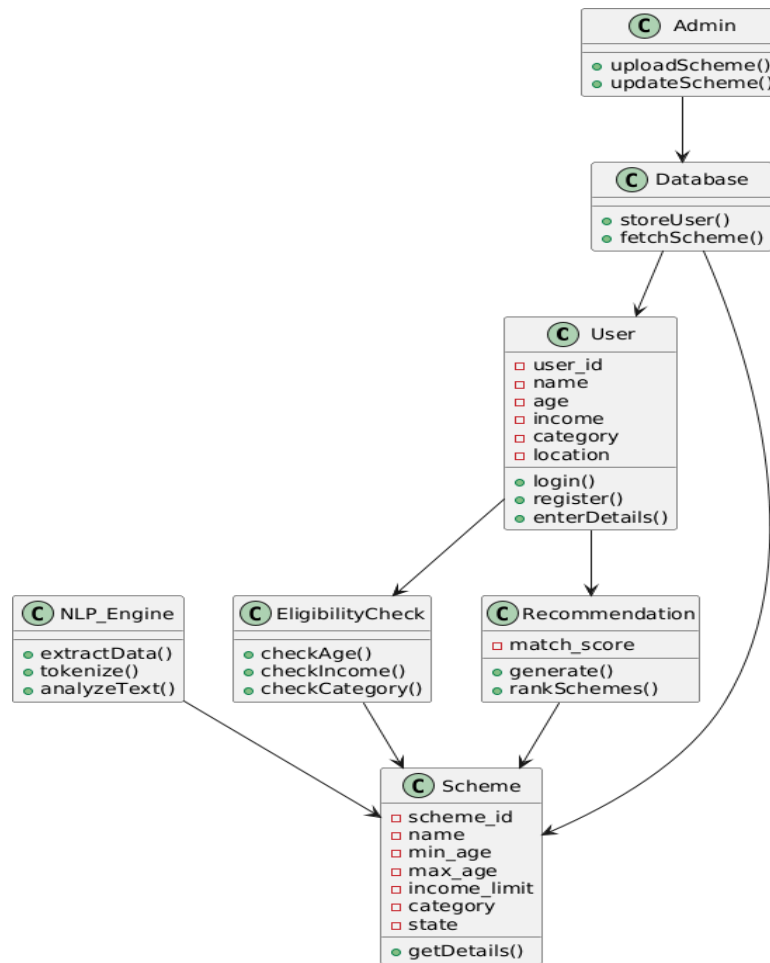


Figure:5.4 Class Diagram of Government Scheme Matching System

The class diagram shows the structure of the system with classes like User, Scheme, NLP Engine, EligibilityCheck, Recommendation, Database, and Admin. The User provides details, the NLP Engine processes scheme data, and the Eligibility Check verifies conditions. The Recommendation class suggests suitable schemes, while the Database stores data and the Admin manages scheme information. All classes work together to provide accurate scheme recommendations

Chapter 5

5.5 Sequence Diagram

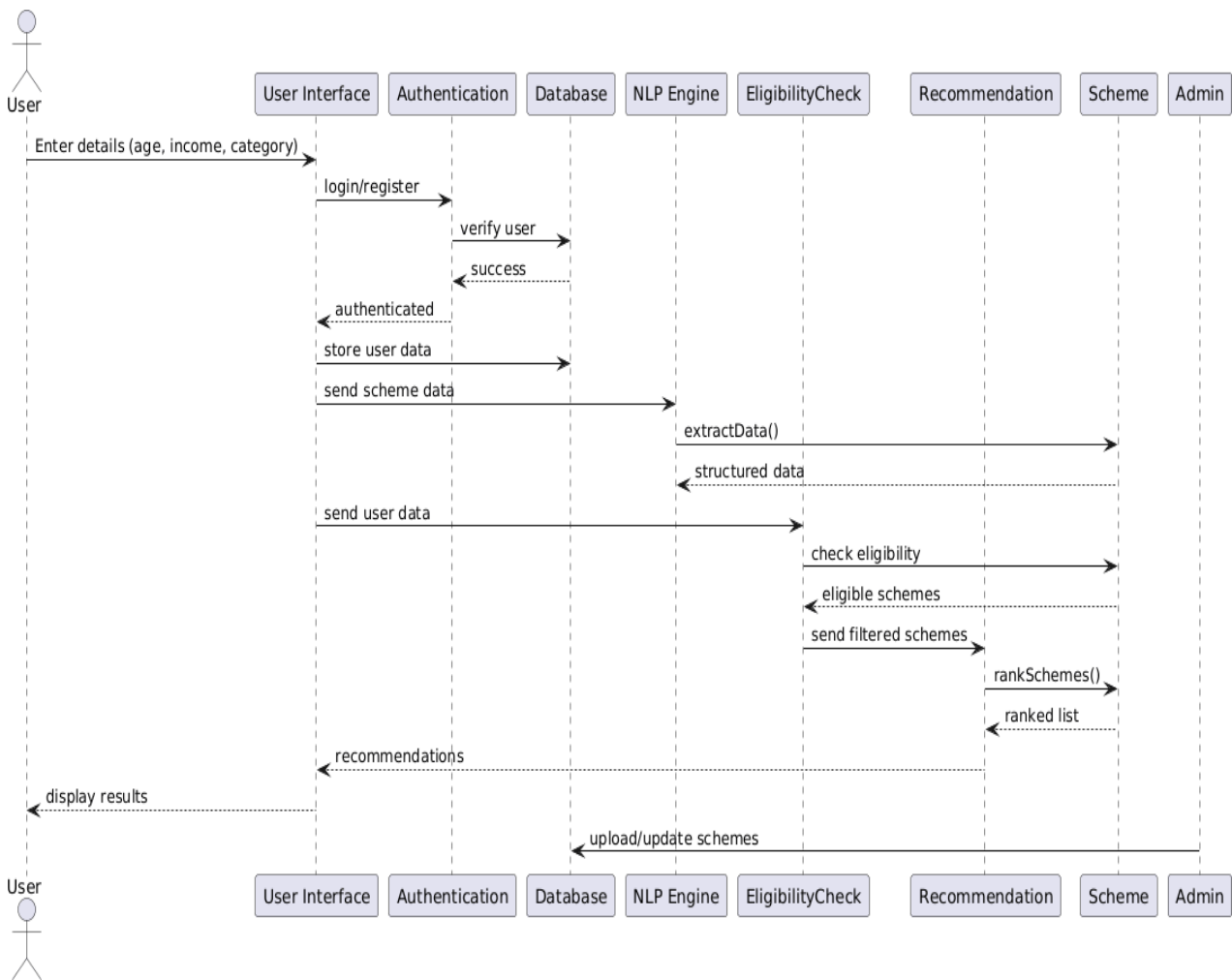


Figure 5.5: Sequence Diagram of Government Scheme Matching System

The sequence diagram illustrates the step-by-step interaction between the user and system components. The process begins when the user enters personal details through the user interface and performs login or registration. The authentication module verifies the user with the database and confirms access

Chapter 5

The user data is then stored in the database, and scheme data is sent to the NLP Engine, which extracts and converts unstructured information into structured data. The eligibility checking module compares user details with scheme criteria and filters eligible schemes.

The recommendation module ranks the schemes based on relevance and generates personalized suggestions. Finally, the results are displayed to the user. The admin can also upload or update scheme data in the database.

5.6 Implementation Details

The implementation of the **Public Welfare Scheme Matching System using NLP** is carried out using a modular architecture consisting of frontend, backend, database, and an NLP processing engine.

1. Frontend Module

The frontend is developed using a web-based interface that allows users to:

- Register/Login
- Enter personal details (age, income, category, location)
- View recommended schemes and explanations

It provides a simple and user-friendly interface for easy interaction.

2.Backend Module

The backend handles core logic and communication between modules. It is responsible for:

- Processing user requests
- Managing APIs
- Handling authentication
- Integrating NLP and recommendation modules

Chapter 5

2. NLP Engine

The NLP module processes unstructured scheme documents and extracts useful information:

- Age limits (min_age, max_age)
- Income criteria
- Category and location details

It uses techniques like tokenization, keyword extraction, and text analysis to convert raw text into structured data.

3. Eligibility Checking Module

This module compares user details with scheme criteria using rule-based logic:

- Age validation
- Income comparison
- Category matching

Only eligible schemes are selected for further processing.

4. Recommendation Module

The system ranks schemes using a **Match Score** based on:

- User's state
- Category match
- Benefit-related keywords (subsidy, grant, etc.)

This ensures personalized and relevant recommendations.

Chapter 5

5. Database

The system uses two databases:

- **User Database** → stores user information
- **Scheme Database** → stores processed scheme data

6. Admin Module

The admin manages scheme data by:

- Uploading new schemes
- Updating existing scheme details
- Maintaining database accuracy

7. Multilingual Support

The system includes a translation feature that converts scheme information into different languages such as Kannada and Hindi, improving accessibility.

8. System Integration

All modules are integrated to work together smoothly:

- Frontend interacts with backend APIs
- Backend connects with NLP engine and database
- Output is generated in real-time

Chapter 6

Results and Discussion

Public welfare schemes play a crucial role in improving the socio-economic conditions of citizens. As governments introduce various schemes for education, healthcare, employment, and financial assistance, the availability of such schemes has increased significantly. However, due to the large number of schemes and complex eligibility criteria, it becomes difficult for citizens to identify the most suitable schemes. The information is often scattered across multiple platforms and presented in unstructured formats, which makes understanding and accessing these schemes time-consuming and confusing. Many eligible beneficiaries fail to utilize these opportunities due to lack of awareness and proper guidance.

To address this issue, we propose a system for intelligent government scheme matching using Natural Language Processing (NLP). The system analyzes scheme documents, extracts eligibility criteria, and matches them with user details to provide accurate recommendations. It simplifies the process by reducing manual effort and providing clear explanations for each suggested scheme. This application ensures better accessibility, improves user experience, and helps citizens easily find and benefit from relevant government welfare schemes.

Chapter 7

Project Plan and Mapped SDGs

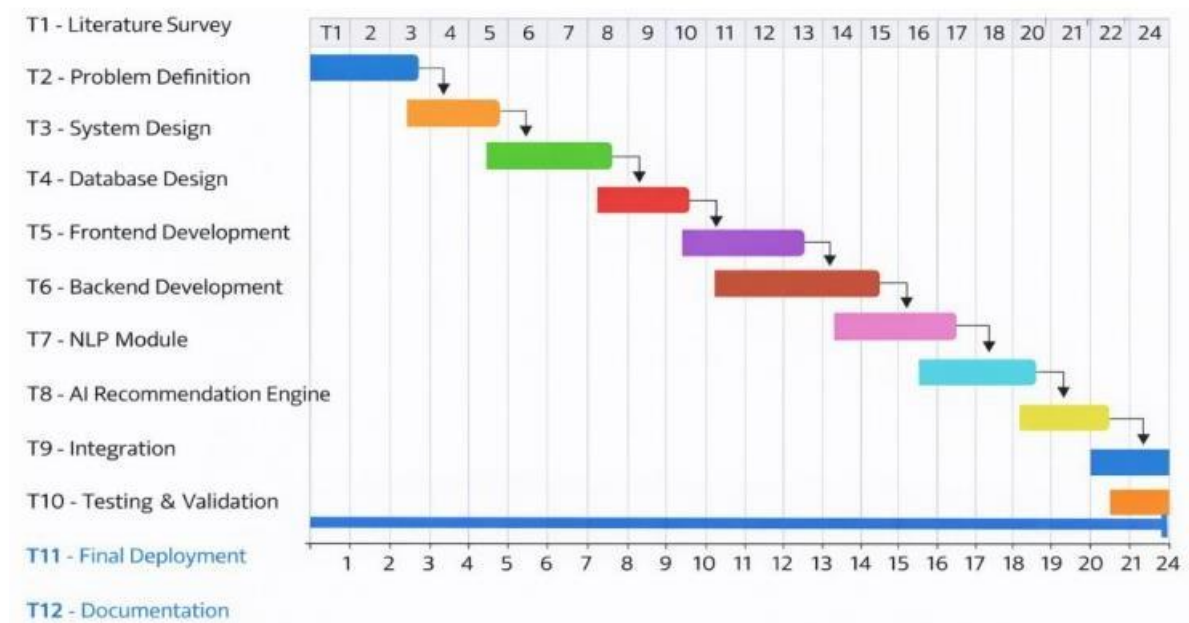


Figure 7.1: Gantt Chart of Public Scheme Matching System

The project “Public Welfare Scheme Matching System using Natural Language Processing (NLP)” is planned over a duration of 24 weeks (approximately 6 months). The work is organized into structured phases including literature survey, problem definition, system design, development, integration, testing, and deployment. Each phase is scheduled in the Gantt chart with proper task dependencies to ensure systematic progress and timely completion of the project.

Mapped SDGs

The project “Public Welfare Scheme Matching System using Natural Language Processing (NLP)” contributes to the following Sustainable Development Goals (SDGs):

Chapter 7

SDG 1: No Poverty

The system helps citizens identify and access suitable government welfare schemes, ensuring that financial and social support reaches economically weaker sections.



Figure 7.2: No Poverty SDG

SDG 9: Industry, Innovation and Infrastructure

The project uses advanced technologies like NLP and AI to develop an innovative and scalable e-governance solution, improving digital infrastructure.



Figure 7.3: Industry, Innovation and Infrastructure

SDG 16: Peace, Justice and Strong Institutions

The system promotes transparency, efficiency, and accessibility in public service delivery by providing clear and explainable scheme recommendations.



Figure 7.3: Industry, Innovation and Infrastructure

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